INDUSTRIAL INTERNSHIP REPORT

QUALITY PREDICTION IN A MINING PROCESS

Project
Machine Learning-Based Prediction of Silica Percentage in Iron-Ore Concentrate

Submitted by

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EXECUTIVE SUMMARY

This report documents the work completed during the upskill Campus industrial internship programme associated with The IoT Academy and UniConverge Technologies Pvt. Ltd. (UCT). The project, titled **"Quality Prediction in a Mining Process,"** applies machine learning to estimate the percentage of silica concentrate in an iron-ore flotation process.

The work uses the dataset *MiningProcess_Flotation_Plant_Database.csv*. The project workflow includes data preparation, time-based aggregation, feature selection, model training and evaluation. The available project notes report 737,453 original rows and 24 columns, with hourly aggregation producing 4,097 records. Random Forest and Gradient Boosting Regressor models were evaluated. Gradient Boosting achieved a reported MAE of 0.5602, RMSE of 0.6993 and R^2 of 0.6249; Random Forest achieved a reported MAE of 0.5867, RMSE of 0.7232 and R^2 of 0.5989.

The completed source code and project documentation were published in the GitHub repository Project10-Mining-Quality-Prediction. This internship provided practical experience in data preprocessing, regression modelling, evaluation metrics, debugging, and documenting a machine-learning workflow. The reported scores are presented as recorded project results and should not be interpreted as a guarantee of performance on every production dataset.

Project at a glance

Item	Details
Project title	Quality Prediction in a Mining Process
Domain	Machine Learning / Industrial Process Analytics
Prediction target	% Silica Concentrate
Main model	GradientBoostingRegressor
Dataset	MiningProcess_Flotation_Plant_Database.csv
Repository	github.com/bhatmouna829-cloud/Project10-Mining-Quality-Prediction

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1. PREFACE

Industrial internships help students connect classroom concepts with practical problems, tools and workflows used in industry. The internship programme facilitated by upskill Campus, The IoT Academy and UCT provided an opportunity to work on a machine-learning problem involving industrial process data.

The assigned project was “Quality Prediction in a Mining Process.” Its aim was to use available process measurements from an iron-ore flotation plant to estimate the percentage of silica concentrate. The work involved understanding the target variable, preparing data for modelling, training regression algorithms and comparing their evaluation metrics.

The project was completed through a structured workflow: inspect the dataset, prepare and aggregate the data, train models, evaluate predictions, review implementation issues and document the final work. The project code and supporting documentation were pushed to GitHub so that the implementation can be reviewed and reproduced.

This work strengthened my understanding of the practical machine-learning lifecycle and the importance of data quality, fair evaluation and clear documentation. It also highlighted that a useful model is not defined only by its algorithm; it must be assessed against the problem, data constraints and intended use.

I am thankful to the internship programme organisers and my institution for the opportunity to learn through this project. I also acknowledge the contribution of educators, documentation authors and learning resources that supported my understanding during implementation.

Message to juniors and peers: approach an unfamiliar project step by step. Understand the data and problem first, test each stage, record what changes, and explain the results honestly. A completed and well-documented project can be a strong learning outcome even when there is scope to improve the model further.

2. INTRODUCTION

Machine learning is increasingly used to analyse industrial data, identify patterns and support process monitoring. In manufacturing and mineral processing, process measurements can vary over time, and changes in operating conditions may be associated with changes in product quality. A regression model can learn relationships between recorded input variables and a numerical quality measure.

The internship project explores this idea through an iron-ore flotation dataset. The model estimates the percentage of silica concentrate from available process variables. This is a prediction task: the model learns from historical observations and produces an estimate for a given set of input measurements. The prediction can support analysis, but it does not replace process expertise or laboratory validation.

2.1 About UniConverge Technologies Pvt. Ltd.

The supplied internship sample describes UniConverge Technologies Pvt. Ltd. as a company established in 2013 and working in digital transformation, with a focus on industrial solutions, sustainability and return on investment. It identifies technologies used across its solutions such as Internet of Things (IoT), cybersecurity, cloud computing, machine learning, communication technologies, Java, Python and frontend development.

The sample also describes UCT offerings in IoT platforms, smart-factory solutions, LoRaWAN-based applications and predictive maintenance. These areas illustrate how data collection, monitoring and analytics can be applied to industrial environments.

2.2 About upskill Campus and The IoT Academy

According to the supplied sample, upskill Campus worked with The IoT Academy and UCT to facilitate the internship process. The IoT Academy is described as the EdTech division of UCT, delivering executive certification programmes in collaboration with academic partners. The programme gave students a structured opportunity to work on a defined project and submit implementation work with a report.

2.3 Internship Objectives

- Gain practical experience in applying machine-learning concepts to an industry-oriented problem.
- Understand the dataset, its features, target variable and preparation requirements.
- Build and compare regression models for estimating silica concentrate percentage.
- Use appropriate regression metrics to evaluate and compare model performance.
- Practise debugging, testing, source-code organisation and technical documentation.
- Publish the completed implementation and report materials in a GitHub repository.

2.4 References

[1] Project repository: https://github.com/bhatmouna829-cloud/Project10-Mining-Quality-Prediction

[2] Dataset file used by the project: MiningProcess_Flotation_Plant_Database.csv. The original external dataset URL was not available in the project details used to prepare this report.

[3] Sample internship report supplied for the report structure: Sample_InternshipReport_USC_UCT.docx.

2.5 Glossary

Term	Meaning
ML	Machine Learning: methods that learn patterns from data.
Regression	A modelling task that predicts a numerical value.
Target	The value the model is trained to estimate; here, % Silica Concentrate.
Feature	An input variable used by a model for prediction.
MAE	Mean Absolute Error; average absolute difference between actual and predicted values.
RMSE	Root Mean Squared Error; penalises larger errors more strongly than MAE.
R^2	Coefficient of determination; indicates how much variation is explained by the model relative to a baseline.
Aggregation	Combining observations over a time interval, such as an hour.

3. PROBLEM STATEMENT

Industrial flotation processes generate measurements from multiple operating variables. Understanding how these variables relate to product quality can be difficult when the dataset is large and observations are recorded over time. The project focuses on predicting the percentage of silica concentrate in iron-ore concentrate using historical flotation-process data.

The dataset used in the project is named *MiningProcess_Flotation_Plant_Database.csv*. Project notes record 737,453 rows and 24 columns in the original dataset. For the implemented hourly modelling workflow, aggregation produced 4,097 records. The prediction target is the “% Silica Concentrate” field.

3.1 Problem definition

Given process measurements for a recorded time period, build a regression model that estimates the silica percentage. The model should be trained on historical data and evaluated using numerical metrics that quantify prediction error and goodness of fit.

3.2 Key challenges

- The raw dataset is large and requires careful inspection and preparation before modelling.
- Process observations are time-based, so aggregation and ordering must be handled consistently.
- Different input variables may have different scales, distributions and relationships with the target.
- A model can appear useful while still producing substantial prediction errors; multiple metrics are needed.
- Any evaluation can be misleading if training and test data are not separated appropriately or if information from the future leaks into training.

3.3 Scope

The project scope covers data preparation, hourly aggregation, regression-model implementation and performance comparison. It is a data-driven prediction prototype. The report does not claim that the model has been deployed in a live mining plant, connected to industrial control systems or validated for operational decision-making.

Expected outcome

A documented machine-learning workflow and a comparison of model metrics, with Gradient Boosting Regressor as the best-performing model among the two reported baseline models.

4. EXISTING AND PROPOSED SOLUTION

4.1 Existing approach and limitations

A basic approach to estimating process quality may rely on manual review of measurements, historical averages or fixed rules. Such approaches can be easy to understand but may not capture complex relationships among several process variables. They can also require repeated manual analysis when the dataset is large.

Approach	Potential limitation
Manual inspection of records	Time-consuming for large datasets and difficult to apply consistently.
Simple average or fixed threshold	May not represent nonlinear relationships between operating variables and quality.
Single model without comparison	Does not show whether another model provides lower prediction error.

These are general limitations of simple baseline approaches; this report does not claim that a particular commercial system was tested or benchmarked.

4.2 Proposed solution

The proposed solution uses supervised regression on historical flotation data. The data is inspected and prepared, observations are aggregated by hour for the main workflow, and two regression algorithms are trained and compared. Evaluation metrics are used to select the stronger model among the implemented alternatives.

- Prepare the dataset and convert or aggregate time-based observations into hourly records.
- Use process measurements as input features and % Silica Concentrate as the prediction target.
- Train a Random Forest regression model as a baseline.
- Train a Gradient Boosting Regressor and compare the results with the baseline.
- Record MAE, RMSE and R^2 for model evaluation.
- Maintain source code, project documentation and implementation notes in GitHub.

4.3 Value addition

The project demonstrates a repeatable data-science workflow and provides a quantitative comparison between two regression approaches. Hourly aggregation reduces the number of records used in the main modelling workflow, while the use of multiple metrics gives a more balanced view of prediction quality than a single score alone.

4.4 Code submission (GitHub link)

Repository: https://github.com/bhatmouna829-cloud/Project10-Mining-Quality-Prediction

The repository contains project code and documentation, including scripts for data preparation and model experiments. The report is based on the project details available at the time of writing; any later code or metric changes should be reflected in the final submitted version.

5. PROPOSED DESIGN / MODEL

The system is organised as a sequence of data and modelling stages. Raw process observations are prepared and aggregated, relevant columns are separated into features and target, regression models are trained, and the trained models are evaluated. The output is a predicted silica concentrate percentage and a set of metrics used to compare models.

5.1 High-level design

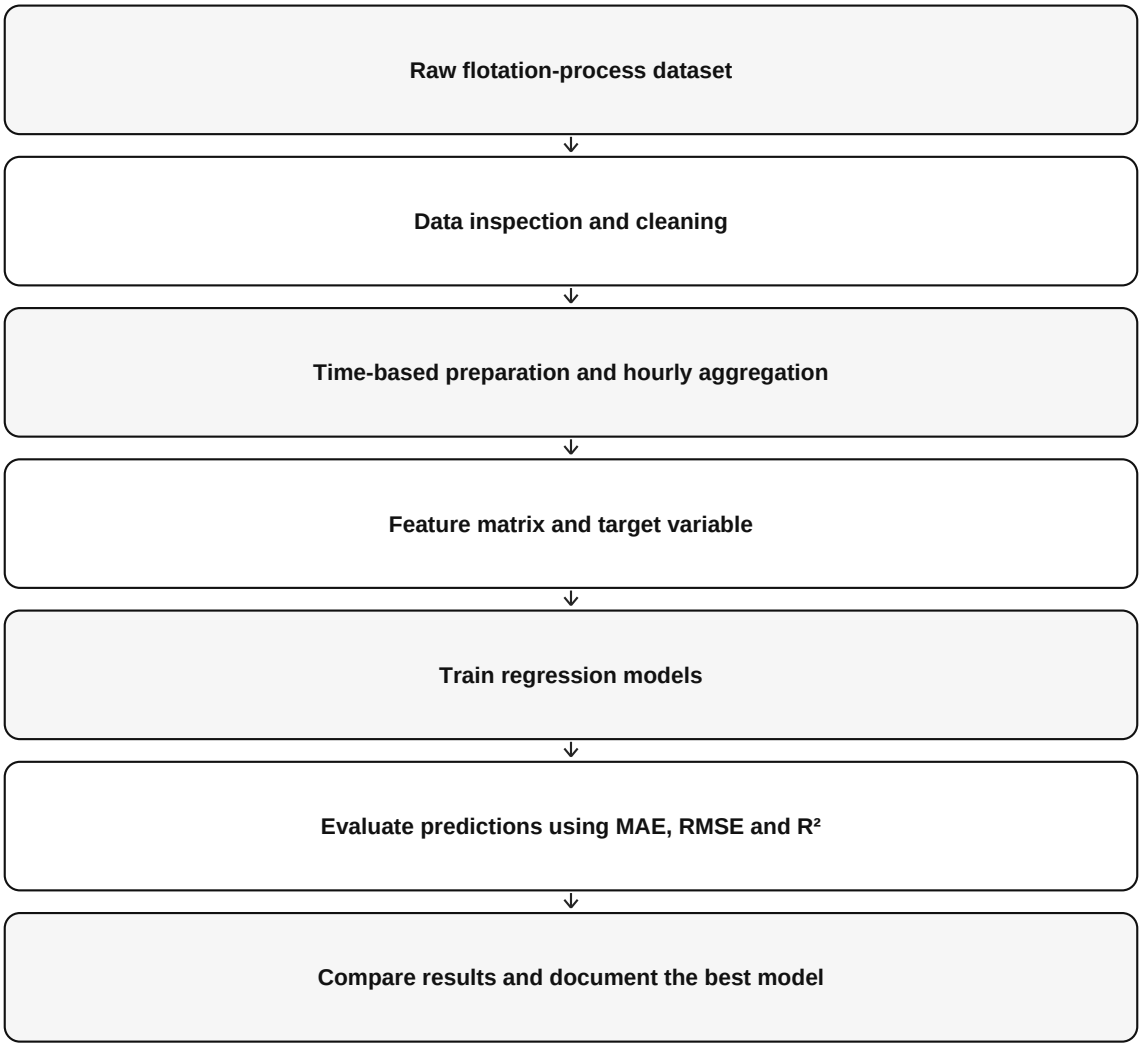


Figure 1. High-level workflow of the quality-prediction system.

5.2 Dataset and features

The project uses process variables recorded in the flotation dataset. The available project notes identify variables such as timestamp, iron and silica feed measurements, starch and amina flow, ore-pulp measurements, flotation air and level measurements, and % Iron Concentrate. The prediction target is % Silica Concentrate.

Data element	Role in modelling
Timestamp	Supports ordering and hourly aggregation.
Process measurements	Input features representing feed, reagent, pulp and flotation conditions.
% Iron Concentrate	One of the recorded process/product variables identified in the project notes; used as an input where included by the script.

Data element	Role in modelling
% Silica Concentrate	Target variable to be predicted.

The exact final feature list can differ by experiment script. The repository scripts include additional experiments such as a model without iron-related inputs and a minute-level model; results in this report focus on the main hourly Random Forest and Gradient Boosting comparison.

5. PROPOSED DESIGN / MODEL (CONTINUED)

5.3 Model design

Random Forest Regression

Random Forest combines predictions from multiple decision trees. In regression, the predictions of individual trees are aggregated to produce a final numerical estimate. It is useful as a baseline because it can model nonlinear relationships and interactions among features without requiring a linear relationship between every input and the target.

Gradient Boosting Regressor

Gradient Boosting builds an additive sequence of weak prediction models, with later models attempting to reduce the errors made by the current ensemble. It can represent complex relationships, but its performance depends on appropriate data preparation and parameter choices. In the reported project comparison, Gradient Boosting achieved lower MAE and RMSE and a higher R^2 than Random Forest.

5.4 Data and model workflow

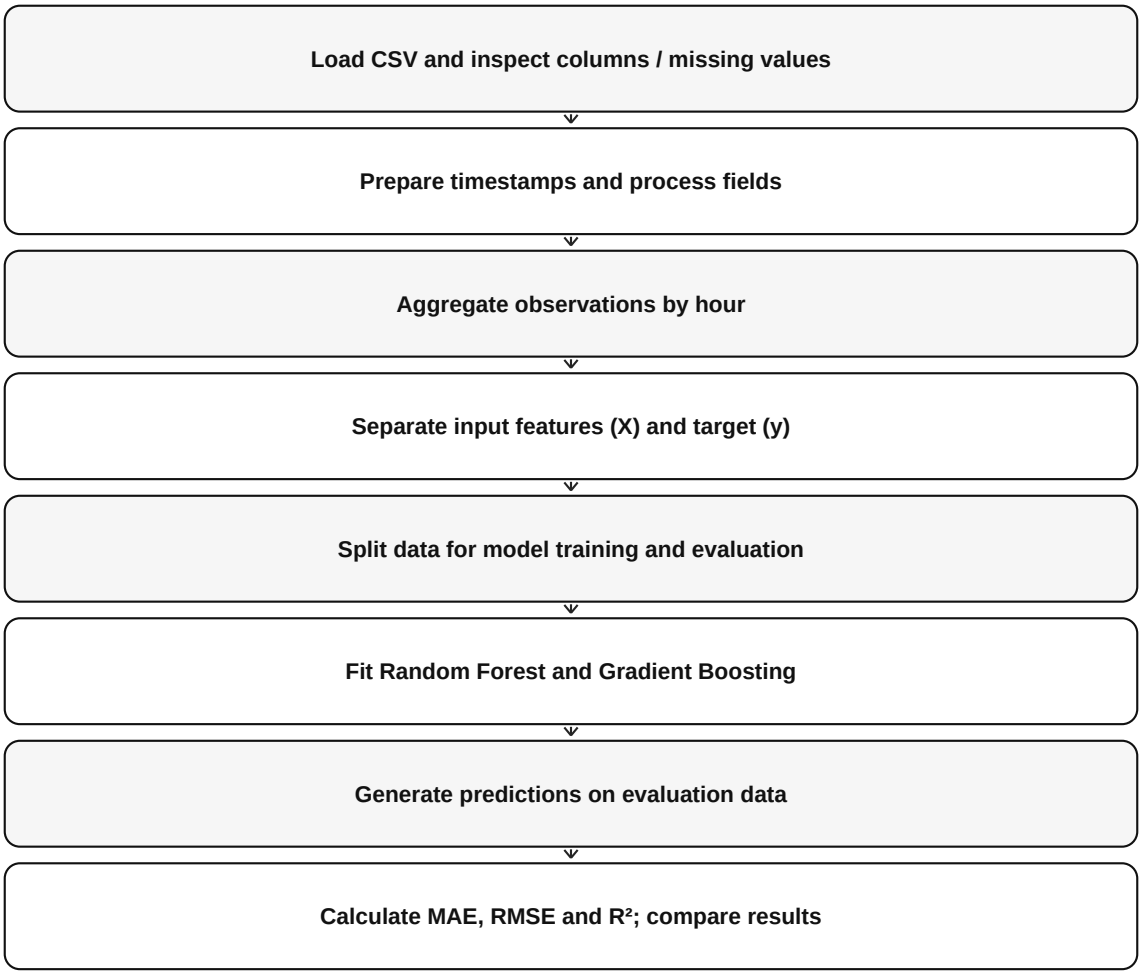


Figure 2. Detailed data and model workflow. The precise split configuration and preprocessing choices should be checked against the final code before a formal reproducibility audit.

5.5 Interfaces and outputs

This project is a Python-based machine-learning implementation rather than a deployed web interface. Its main interfaces are the data files, Python scripts and generated evaluation outputs. The repository includes scripts for cleaning and aggregation and for running model experiments. The primary output is a numerical estimate of silica concentrate percentage, accompanied by model-performance metrics.

- Input: flotation-process dataset in CSV format.
- Processing: data preparation, aggregation, feature/target separation and model fitting.
- Output: predicted target values and evaluation metrics for model comparison.
- Documentation: README and project report files maintained in GitHub.

6. PERFORMANCE TEST

Model evaluation is necessary to determine how closely predictions match observed values. This project reports three regression metrics: Mean Absolute Error (MAE), Root Mean Squared Error (RMSE) and coefficient of determination (R^2). Lower MAE and RMSE indicate smaller errors; a higher R^2 indicates that the model explains more variation relative to a baseline.

6.1 Test plan / test cases

Test ID	Test / check	Expected outcome
T1	Load the source CSV and inspect the dataset shape.	Dataset loads and expected columns are available.
T2	Run the data preparation and hourly aggregation workflow.	Hourly dataset is produced; project notes report 4,097 records.
T3	Train the Random Forest regression model.	Model produces numerical predictions for evaluation.
T4	Train the Gradient Boosting Regressor.	Model produces numerical predictions for evaluation.
T5	Calculate MAE, RMSE and R^2 for each model.	Comparable evaluation metrics are recorded.
T6	Compare model metrics and document the outcome.	The model with lower error and stronger fit is identified among tested models.

6.2 Test procedure

1. Load the dataset and inspect the number of records, columns and data types.
2. Prepare the time field and create the hourly aggregated dataset used by the main experiment.
3. Select the input columns and the % Silica Concentrate target according to the experiment script.
4. Train the Random Forest and Gradient Boosting regression models using the implemented training workflow.
5. Generate predictions for the evaluation data and calculate MAE, RMSE and R^2.
6. Compare the scores and record the better-performing model for the reported experiment.

6.3 Performance outcome

Model	MAE ↓	RMSE ↓	R^2 ↑
Random Forest Regressor	0.5867	0.7232	0.5989
Gradient Boosting Regressor	0.5602	0.6993	0.6249

Interpretation

For the reported comparison, Gradient Boosting has a lower MAE by 0.0265 and a lower RMSE by 0.0239 than Random Forest. Its R^2 is higher by 0.0260. Therefore, Gradient Boosting is the stronger of these two models on the reported evaluation. The results indicate useful predictive signal, while also showing that a meaningful portion of target variation remains unexplained.

Limitations of the test

The metrics are reproduced from the available project notes. This report does not independently rerun the code or verify the train/test split, data leakage controls, confidence intervals or performance on new plant data. For operational use, evaluation should be repeated with a documented split strategy, time-aware validation where appropriate, and external validation on data collected under different operating conditions.

7. MY LEARNINGS

The project helped me understand how a machine-learning problem is developed from a raw dataset to an evaluated model. It also reinforced that the quality of the data-preparation pipeline is as important as the choice of algorithm.

Technical learning

- Understood how to frame an industrial quality-estimation problem as a supervised regression task.
- Practised inspecting a large CSV dataset and identifying input features and a numerical target.
- Learned why timestamps and aggregation intervals matter when working with time-based process data.
- Compared Random Forest and Gradient Boosting regression models.
- Improved my understanding of MAE, RMSE and R^2 and how they complement one another.
- Learned to organise scripts and documentation so the project can be reviewed through a version-controlled repository.

Problem-solving and work habits

- Worked through implementation issues by checking the workflow one stage at a time.
- Tested data preparation and model steps separately rather than treating the project as a single operation.
- Reviewed output values and evaluation scores to check whether the implementation behaved as expected.
- Recognised the importance of recording assumptions and not overstating model performance.

Career relevance

This internship project supports my development in machine learning and applied data science. It gives me a concrete example of data preparation, regression modelling, evaluation and project documentation that I can discuss in technical interviews. It also motivates me to deepen my knowledge of model validation, feature engineering, experiment tracking and deployment practices.

Overall reflection

The project was a valuable practical exercise because it required more than writing a model-training statement: the data had to be understood, prepared, tested and documented. Completing the repository and report gave me experience in presenting technical work in a structured form. The next step is to improve validation and make the experimental setup more reproducible.

8. FUTURE WORK SCOPE

The project provides a useful baseline for exploring machine-learning-based quality prediction. Several improvements can be considered before the solution is used for real industrial decisions.

8.1 Data and validation improvements

- Document missing-value handling, duplicate checks, outlier treatment and the final list of model features.
- Record the exact training, validation and test split and use time-aware validation if observations are temporally dependent.
- Evaluate the model on a separate period or new batch of plant data to assess generalisation.
- Check for data leakage, especially when aggregating time-based measurements or selecting features.
- Report variability across repeated runs or folds, not only a single score.

8.2 Model improvements

- Tune model hyperparameters systematically and compare results using the same evaluation procedure.
- Compare additional regression algorithms where appropriate, including linear baselines and other tree-based methods.
- Study feature importance or explainability methods to understand which process variables are associated with predictions.
- Analyse residuals to identify time periods or operating conditions where errors are higher.
- Compare the hourly model with alternative aggregation intervals only when the evaluation setup remains fair.

8.3 Deployment and monitoring

- Develop a simple dashboard to display predictions, actual values and recent error metrics.
- Add input validation and clear warnings when required process measurements are missing or outside expected ranges.
- Monitor model drift and retrain only when justified by new data and documented evaluation.
- Keep a human expert in the decision loop; predictions should support, not automatically replace, process decisions.
- Establish access control, versioning and logging before connecting the model to any operational system.

Conclusion

The internship project, “Quality Prediction in a Mining Process,” applied regression modelling to estimate % Silica Concentrate from flotation-process data. The reported comparison favoured Gradient Boosting over Random Forest on MAE, RMSE and R^2. The work strengthened practical skills in data preparation, model evaluation, debugging and documentation. Further validation, reproducibility checks and deployment safeguards would be necessary before using the model in a live industrial setting.

— End of Report —